

Supplementary materials:

COGS: A Causal Representation Learning Framework for Out-of-Distribution Generalization in Time Series

A Proof and Deviations

A.1 Convergence Analysis

Let $\mathcal{L}_{\text{pred}} = \mathcal{L}_{\text{TCRL}} + \sum_{d \in \mathcal{E}_{\text{train}}} \mathcal{L}_{\text{CE}}^{(d)} + \beta \cdot \text{Var}_{d \in \mathcal{E}_{\text{train}}} [\mathcal{L}_{\text{CE}}^{(d)}]$ and $\mathcal{L}_{\text{cluster}} = \mathcal{L}_{\text{proto}} + \eta \cdot \mathcal{L}_{\text{con}} + \gamma \cdot \mathcal{L}_{\text{ind}}$. We show in Theorem 1 that under certain assumptions, through the alternating optimization of the causal invariance prediction phase and the domain inference phase, the algorithm will eventually converge and the causal graph will tend to be stable.

Assumption 1 *Let $\hat{\mathbf{M}} = 1 - \mathbf{M}$. For any latent variables $Z^i \in \mathbb{R}^T, i = 1, \dots, N$, if Z^i is a causal variable, then in Causal Invariant Prediction Stage, its removal is detrimental to causal invariant prediction. But in Domain Inference Stage, its inclusion will introduce harmful noise for domain inference, that is, $\exists \lambda_1 > 0$ and $\lambda_2 > 0$, :*

$$\mathcal{L}_{\text{pred}}(Y, \mathbf{X}, \mathbf{Z} \odot \mathbf{M}_{i=0}) - \mathcal{L}_{\text{pred}}(Y, \mathbf{X}, \mathbf{Z} \odot \mathbf{M}_{i=1}) > \lambda_1, \quad (1)$$

$$\mathcal{L}_{\text{cluster}}(Y, \mathbf{X}, \mathbf{Z} \odot \hat{\mathbf{M}}_{i=1}) - \mathcal{L}_{\text{cluster}}(Y, \mathbf{X}, \mathbf{Z} \odot \hat{\mathbf{M}}_{i=0}) > \lambda_2. \quad (2)$$

If Z^i is a non-causal variable, then in Causal Invariant Prediction Stage, its inclusion provides no predictive benefit for causal invariant prediction. But in Domain Inference Stage, its removal is detrimental to domain inference, that is:

$$\lambda_1 > \mathcal{L}_{\text{pred}}(Y, \mathbf{X}, \mathbf{Z} \odot \mathbf{M}_{i=1}) - \mathcal{L}_{\text{pred}}(Y, \mathbf{X}, \mathbf{Z} \odot \mathbf{M}_{i=0}) \geq 0, \quad (3)$$

$$\lambda_2 > \mathcal{L}_{\text{cluster}}(Y, \mathbf{X}, \mathbf{Z} \odot \hat{\mathbf{M}}_{i=0}) - \mathcal{L}_{\text{cluster}}(Y, \mathbf{X}, \mathbf{Z} \odot \hat{\mathbf{M}}_{i=1}) \geq 0. \quad (4)$$

where $\mathbf{M}_{i=l}$ is the sampled binary mask with element $M_{i,1:T} = l$.

Theorem 1 *Given a time-series dataset $(\mathbf{X}, Y)$, we can identify the latent variables $Z^i \in \mathbb{R}^T, i = 1, 2, \dots, N$ from observed $\mathbf{X}$ through the temporal causal representation learning module. Under Assumption 1, $\exists \lambda$, the element of causal probability graph $\mathcal{G}_i = \sigma(\theta_i)$ converges to 0 if latent variable Z^i does not causally related to Y , and $\mathcal{G}_i$ converges to 1 if latent variable Z^i is the direct cause of Y .*

The implications behind Assumption 1 can be intuitively explained. It means there exists a threshold λ_1 to binarize $\mathcal{L}_{\text{pred}}(Y, \mathbf{Z} \odot \mathbf{M}_{i=0}) - \mathcal{L}_{\text{pred}}(Y, \mathbf{Z} \odot \mathbf{M}_{i=1})$, serving as an indicator as to whether latent variable Z^i contributes to prediction of Y . There also exists a threshold λ_2 to binarize $\mathcal{L}_{\text{cluster}}(Y, \mathbf{X}, \mathbf{Z} \odot \hat{\mathbf{M}}_{i=1}) - \mathcal{L}_{\text{cluster}}(Y, \mathbf{X}, \mathbf{Z} \odot \hat{\mathbf{M}}_{i=0})$, serving as an indicator as to whether latent variable Z^i contributes to prediction of d .

Proof of Theorem 1: We analyze the sign of the gradient for a single parameter θ_i in each stage. The gradient of the expected total loss is derived using the REINFORCE trick [1].

(1). In *Causal Invariant Prediction Stage*, the loss function is:

$$\mathcal{L}_{\text{stage1}}(\mathbf{X}, Y, \Theta) = \mathcal{L}_{\text{pred}}(Y, \mathbf{X}, \mathbf{Z} \odot \mathbf{M}) + \lambda \|\sigma(\Theta)\|_1.$$

Next, we omit the time dimension of $\mathbf{M}$ since all time points for the same variable share the same probability. Let $\mathbf{M}_i$ denote the mask value of the latent variable Z^i , $\mathbf{M}_i \sim \text{Ber}(\sigma(\theta_i))$. The gradient of θ_i is calculated as

$$\begin{aligned} \frac{\partial}{\partial \theta_i} \mathbb{E}_{\mathbf{M}_i}[\mathcal{L}_{\text{stage1}}] &= \frac{\partial}{\partial \theta_i} \mathbb{E}_{\mathbf{M}_i}[\mathcal{L}_{\text{pred}}(Y, \mathbf{X}, \mathbf{Z} \odot \mathbf{M})] + \lambda \sigma'(\theta_i) \\ &= \sigma'(\theta_i) (\mathcal{L}_{\text{pred}}(Y, \mathbf{X}, \mathbf{Z} \odot \mathbf{M}_{i=1}) - \mathcal{L}_{\text{pred}}(Y, \mathbf{X}, \mathbf{Z} \odot \mathbf{M}_{i=0}) + \lambda). \end{aligned}$$

If latent variable Z^i is a non-causal variable with respect to Y , according to inequality 3 of Assumption 1, we can derive that

$$\frac{\partial}{\partial \theta_i} \mathbb{E}_{\mathbf{M}_i}[\mathcal{L}_{\text{stage1}}] = \lambda_{\text{stage1}} \sigma'(\theta_i)$$

where $\lambda_{\text{stage1}} = (\mathcal{L}_{\text{pred}}(Y, \mathbf{X}, \mathbf{Z} \odot \mathbf{M}_{i=1}) - \mathcal{L}_{\text{pred}}(Y, \mathbf{X}, \mathbf{Z} \odot \mathbf{M}_{i=0}) + \lambda) > \lambda > 0$. This is a sigmoidal gradient, whose convergence is analyzed in [2]. Since this gradient is always positive, according to the gradient descent update formula: $\theta_i^{k+1} = \theta_i^k - \frac{\partial}{\partial \theta_i} \mathbb{E}_{\mathbf{M}_i}[\mathcal{L}_{\text{stage1}}]$, θ_i will go towards $-\infty$ and $\mathbf{M}_i \rightarrow 0$. Likewise, if latent variable Z^i is a causal variable with respect to Y , according to inequality 1 of Assumption 1, we can derive that:

$$\frac{\partial}{\partial \theta_i} \mathbb{E}_{\mathbf{M}_i}[\mathcal{L}_{\text{stage1}}] = \sigma'(\theta_i) (\mathcal{L}_{\text{pred}}(Y, \mathbf{X}, \mathbf{Z} \odot \mathbf{M}_{i=1}) - \mathcal{L}_{\text{pred}}(Y, \mathbf{X}, \mathbf{Z} \odot \mathbf{M}_{i=0}) + \lambda) < (\lambda - \lambda_1) \sigma'(\theta_i)$$

When we select the hyperparameter $\lambda < \lambda_1$, we have

$$\frac{\partial}{\partial \theta_i} \mathbb{E}_{\mathbf{M}_i}[\mathcal{L}_{\text{stage1}}] < (\lambda - \lambda_1) \sigma'(\theta_i) < 0$$

This gradient is expected to be negative, according to the gradient descent update formula: $\theta_i^{k+1} = \theta_i^k - \frac{\partial}{\partial \theta_i} \mathbb{E}_{\mathbf{M}_i}[\mathcal{L}_{\text{stage1}}]$, θ_i will go towards $+\infty$ and $\mathbf{M}_i \rightarrow 1$.

(2). In *Domain Inference Stage*, the loss function is:

$$\mathcal{L}_{\text{stage2}}(\mathbf{X}, Y, \Theta) = \mathcal{L}_{\text{cluster}}(Y, \mathbf{X}, \mathbf{Z} \odot \hat{\mathbf{M}}) + \lambda \|\sigma(\Theta)\|_1.$$

where $\hat{\mathbf{M}} = 1 - \mathbf{M}$ and $\hat{\mathbf{M}}_i \sim \text{Ber}(1 - \sigma(\theta_i))$. Similiar to *Causal Invariant Prediction Stage*, The gradient of θ_i is calculated as

$$\begin{aligned} \frac{\partial}{\partial \theta_i} \mathbb{E}_{\hat{\mathbf{M}}_i}[\mathcal{L}_{\text{stage2}}] &= \frac{\partial}{\partial \theta_i} \mathbb{E}_{\hat{\mathbf{M}}_i}[\mathcal{L}_{\text{cluster}}(Y, \mathbf{X}, \mathbf{Z} \odot \hat{\mathbf{M}})] + \lambda \sigma'(\theta_i) \\ &= \sigma'(\theta_i) (\mathcal{L}_{\text{cluster}}(Y, \mathbf{X}, \mathbf{Z} \odot \hat{\mathbf{M}}_{i=0}) - \mathcal{L}_{\text{cluster}}(Y, \mathbf{X}, \mathbf{Z} \odot \hat{\mathbf{M}}_{i=1}) + \lambda). \end{aligned}$$

If latent variable Z^i is a non-causal variable with respect to Y , according to inequality 4 of Assumption 1, we can derive that

$$\frac{\partial}{\partial \theta_i} \mathbb{E}_{\hat{\mathbf{M}}_i}[\mathcal{L}_{\text{stage2}}] = \lambda_{\text{stage2}} \sigma'(\theta_i)$$

where $\lambda_{stage2} = (\mathcal{L}_{\text{cluster}}(Y, \mathbf{X}, \mathbf{Z} \odot \hat{\mathbf{M}}_{i=0}) - \mathcal{L}_{\text{cluster}}(Y, \mathbf{X}, \mathbf{Z} \odot \hat{\mathbf{M}}_{i=1}) + \lambda) > \lambda > 0$. Thus θ_i will go towards $-\infty$ and $\mathbf{M}_i \rightarrow 0$ while $\hat{\mathbf{M}}_i \rightarrow 1$. Likewise, if latent variable Z^i is a causal variable with respect to Y , according to inequality 2 of Assumption 1, we can derive that:

$$\frac{\partial}{\partial \theta_i} \mathbb{E}_{\hat{\mathbf{M}}_i}[\mathcal{L}_{\text{stage2}}] = \sigma'(\theta_i)(\mathcal{L}_{\text{cluster}}(Y, \mathbf{X}, \mathbf{Z} \odot \hat{\mathbf{M}}_{i=0}) - \mathcal{L}_{\text{pred}}(Y, \mathbf{X}, \mathbf{Z} \odot \hat{\mathbf{M}}_{i=1}) + \lambda) < (\lambda - \lambda_2)\sigma'(\theta_i)$$

When we select the hyperparameter $\lambda < \lambda_2$, we have

$$\frac{\partial}{\partial \theta_i} \mathbb{E}_{\hat{\mathbf{M}}_i}[\mathcal{L}_{\text{stage2}}] < (\lambda - \lambda_2)\sigma'(\theta_i) < 0$$

This gradient is expected to be negative, according to the gradient descent update formula, θ_i will go towards $+\infty$ and $\mathbf{M}_i \rightarrow 1$ while $\hat{\mathbf{M}}_i \rightarrow 0$.

In summary, both optimization stages provide gradient signals with consistent signs and correct directions. Under alternating optimization, the updates in each stage will strengthen rather than weaken the results of the other stage. Therefore, our method will eventually converge, and the learnable causal mask will converge to the true causal graph that can accurately distinguish causal variables from non-causal variables. In our implementation, we use Gumbel-Softmax estimator for improved performance and lower variance [3].

A.2 Proof of Identifiability Theorem

First of all, we define the identifiability of time-delayed latent causal processes in the representation function space in Definition 1, as proposed by [4].

Definition 1 (Identifiable Latent Causal Processes) *Formally let $\{\mathbf{x}_t\}_{t=1}^T$ be a sequence of observed variables generated by the true temporally causal latent processes specified by $(f_i, \boldsymbol{\theta}_d, p(\epsilon_i), \mathbf{g})$. A learned generative model $(\hat{f}_i, \hat{\boldsymbol{\theta}}_d, \hat{p}(\epsilon_i), \hat{\mathbf{g}})$ is observationally equivalent to $(f_i, \boldsymbol{\theta}_d, p(\epsilon_i), \mathbf{g})$ if the model distribution $p_{\hat{f}_i, \hat{\boldsymbol{\theta}}_d, \hat{p}(\epsilon_i), \hat{\mathbf{g}}}(\{\mathbf{x}_t\}_{t=1}^T)$ matches the data distribution $p_{f_i, \boldsymbol{\theta}_d, p(\epsilon_i), \mathbf{g}}(\{\mathbf{x}_t\}_{t=1}^T)$ everywhere. We say latent causal processes are identifiable if observational equivalence can lead to identifiability of the latent variables up to permutation π and component-wise invertible transformation T :*

$$p_{\hat{f}_i, \hat{\boldsymbol{\theta}}_d, \hat{p}(\epsilon_i), \hat{\mathbf{g}}}(\{\mathbf{x}_t\}_{t=1}^T) = p_{f_i, \boldsymbol{\theta}_d, p(\epsilon_i), \mathbf{g}}(\{\mathbf{x}_t\}_{t=1}^T) \Rightarrow \hat{\mathbf{g}}(x_t) = \mathbf{g} \circ \pi \circ T, \quad \forall \mathbf{x}_t \in \mathcal{X}, \quad (5)$$

where $\mathcal{X}$ is the observation space.

In the following, we give a rigorous mathematical statement of the identifiability theorem and the corresponding proof, which is adapted from [5].

Theorem 2 *Consider the following data generation process: $\mathbf{g}$:*

$$\mathbf{x}_t = \mathbf{g}(\mathbf{z}_t), \quad (6)$$

$$z_{t,i} = h_i(pa(\mathbf{z}_{t,i}), d, \epsilon_{t,i}). \quad (7)$$

Let $\eta_{tk}(d) \triangleq \log p(z_{t,k} | \mathbf{z}_{t-1}, d)$ where $d \in \{1, \dots, M\}$ and assume that $\eta_{tk}(d)$ is twice differentiable in $z_{t,k}$ and is differentiable in $z_{t-1,l}, l = 1, 2, \dots, n$. Suppose that there exists an invertible function $\hat{\mathbf{g}}$

that maps $\mathbf{x}_t$ to $\hat{\mathbf{z}}_t$, i.e., $\hat{\mathbf{z}}_t = \hat{\mathbf{g}}(\mathbf{x}_t)$ such that the components of $\hat{\mathbf{z}}_t$ are mutually independent conditional on $\hat{\mathbf{z}}_{t-1}$. Let

$$\mathbf{v}_{t,k}(d) \triangleq \left(\frac{\partial^2 \eta_{tk}(d)}{\partial z_{t,k} \partial z_{t-1,1}}, \dots, \frac{\partial^2 \eta_{tk}(d)}{\partial z_{t,k} \partial z_{t-1,N}} \right)^\top, \quad (8)$$

$$\dot{\mathbf{v}}_{t,k}(u) \triangleq \left(\frac{\partial^3 \eta_{tk}(d)}{\partial z_{t,k}^2 \partial z_{t-1,1}}, \dots, \frac{\partial^3 \eta_{tk}(d)}{\partial z_{t,k}^2 \partial z_{t-1,N}} \right)^\top, \quad (9)$$

$$\mathbf{s}_{t,k} \triangleq (\mathbf{v}_{t,k}(1)^\top, \dots, \mathbf{v}_{t,k}(M)^\top, \Delta^2(2), \dots, \Delta^2(M))^\top, \quad (10)$$

$$\dot{\mathbf{s}}_{t,k} \triangleq (\dot{\mathbf{v}}_{t,k}(1)^\top, \dots, \dot{\mathbf{v}}_{t,k}(M)^\top, \Delta(2), \dots, \Delta(M))^\top. \quad (11)$$

where $\Delta^2(d) = \frac{\partial^2 \eta_{tk}(d)}{\partial z_{t,k}^2} - \frac{\partial^2 \eta_{tk}(d-1)}{\partial z_{t,k}^2}$ and $\Delta(d) = \frac{\partial \eta_{tk}(d)}{\partial z_{t,k}} - \frac{\partial \eta_{tk}(d-1)}{\partial z_{t,k}}$. Suppose $\mathbf{x}_t = \mathbf{g}(\mathbf{z}_t)$ and that the conditional distribution $p(z_{t,k} | \mathbf{z}_{t-1})$ may change across M domains. Suppose that the components of $\mathbf{z}_t$ are mutually independent conditional on $\mathbf{z}_{t-1}$ in each context. Assume that the components of $\hat{\mathbf{z}}_t$ produced by $\hat{\mathbf{g}}$ are also mutually independent conditional on $\hat{\mathbf{z}}_{t-1}$. If the $2n$ function vectors $\mathbf{s}_{t,k}$ and $\dot{\mathbf{s}}_{t,k}$, with $k = 1, 2, \dots, n$, are linearly independent, then $\hat{\mathbf{z}}_t$ is a permuted invertible component-wise transformation of $\mathbf{z}_t$.

Proof of Theorem 2: From $\mathbf{x}_t = \mathbf{g}(\mathbf{z}_t)$ and $\hat{\mathbf{z}}_t = \hat{\mathbf{g}}(\mathbf{x}_t)$ we can deduce that $\mathbf{z}_t = (\mathbf{g}^{-1} \circ \hat{\mathbf{g}})(\hat{\mathbf{z}}_t) = \mathbf{h}(\hat{\mathbf{z}}_t)$, where $\mathbf{h} \triangleq \mathbf{g}^{-1} \circ \hat{\mathbf{g}}$. Since both $\mathbf{g}$ and $\hat{\mathbf{g}}$ are invertible, $\mathbf{h}$ is invertible. Let $\mathbf{H}_t$ be the Jacobian matrix of the transformation $\mathbf{h}(\hat{\mathbf{z}}_t)$, and denote by $\mathbf{H}_{kit}$ its (k, i) th entry.

First, it is straightforward to see that if the components of $\hat{\mathbf{z}}_t$ are mutually independent conditional on previous $\hat{\mathbf{z}}_{t-1}$ and current d , then for any $i \neq j$, $\hat{z}_{ti}$ and $\hat{z}_{tj}$ are conditionally independent given $\hat{\mathbf{z}}_{t-1} \cup (\hat{\mathbf{z}}_t \setminus \{\hat{z}_{ti}, \hat{z}_{tj}\}) \cup \{d\}$. Mutual independence of the components of $\hat{\mathbf{z}}_t$ conditional on $\hat{\mathbf{z}}_{t-1}$ and d implies that $\hat{z}_{ti}$ is independent from $\hat{\mathbf{z}}_t \setminus \{\hat{z}_{ti}, \hat{z}_{tj}\}$ conditional on $\hat{\mathbf{z}}_{t-1}$ and d , i.e.,

$$p(\hat{z}_{ti} | \hat{\mathbf{z}}_{t-1}, d) = p(\hat{z}_{ti} | \hat{\mathbf{z}}_{t-1} \cup (\hat{\mathbf{z}}_t \setminus \{\hat{z}_{ti}, \hat{z}_{tj}\}), d). \quad (12)$$

At the same time, it also implies $\hat{z}_{ti}$ is independent from $\hat{\mathbf{z}}_t \setminus \{\hat{z}_{ti}\}$ conditional on $\hat{\mathbf{z}}_{t-1}$ and d , i.e.,

$$p(\hat{z}_{ti} | \hat{\mathbf{z}}_{t-1}, d) = p(\hat{z}_{ti} | \hat{\mathbf{z}}_{t-1} \cup (\hat{\mathbf{z}}_t \setminus \{\hat{z}_{ti}\}), d). \quad (13)$$

Combining the above two equations gives

$$p(\hat{z}_{ti} | \hat{\mathbf{z}}_{t-1} \cup (\hat{\mathbf{z}}_t \setminus \{\hat{z}_{ti}\}), d) = p(\hat{z}_{ti} | \hat{\mathbf{z}}_{t-1} \cup (\hat{\mathbf{z}}_t \setminus \{\hat{z}_{ti}, \hat{z}_{tj}\}) \cup \{d\}), \quad (14)$$

i.e., for $i \neq j$, $\hat{z}_{ti}$ and $\hat{z}_{tj}$ are conditionally independent given $\hat{\mathbf{z}}_{t-1} \cup (\hat{\mathbf{z}}_t \setminus \{\hat{z}_{ti}, \hat{z}_{tj}\}) \cup \{d\}$.

We then make use of the fact that if $\hat{z}_{ti}$ and $\hat{z}_{tj}$ are conditionally independent given $\hat{\mathbf{z}}_{t-1} \cup (\hat{\mathbf{z}}_t \setminus \{\hat{z}_{ti}, \hat{z}_{tj}\}) \cup \{d\}$, then

$$\frac{\partial^2 \log p(\hat{\mathbf{z}}_t, \hat{\mathbf{z}}_{t-1}, d)}{\partial \hat{z}_{ti} \partial \hat{z}_{tj}} = 0, \quad (15)$$

assuming the cross second-order derivative exists. Since $p(\hat{\mathbf{z}}_t, \hat{\mathbf{z}}_{t-1}, d) = p(\hat{\mathbf{z}}_t | \hat{\mathbf{z}}_{t-1}, d)p(\hat{\mathbf{z}}_{t-1}, d)$ while $p(\hat{\mathbf{z}}_{t-1}, d)$ does not involve $\hat{z}_{ti}$ or $\hat{z}_{tj}$, the above equality is equivalent to

$$\frac{\partial^2 \log p(\hat{\mathbf{z}}_t | \hat{\mathbf{z}}_{t-1}, d)}{\partial \hat{z}_{ti} \partial \hat{z}_{tj}} = 0. \quad (16)$$

Then for any domain d , the Jacobian matrix of the mapping from $(\mathbf{x}_{t-1}, \hat{\mathbf{z}}_t)$ to $(\mathbf{x}_{t-1}, \mathbf{z}_t)$ is $\begin{bmatrix} \mathbf{I} & \mathbf{0} \\ * & \mathbf{H}_t \end{bmatrix}$, where $*$ stands for a matrix, and the absolute value of the determinant of this Jacobian matrix is $|\mathbf{H}_t|$. Therefore $p(\hat{\mathbf{z}}_t, \mathbf{x}_{t-1} \mid d) = p(\mathbf{z}_t, \mathbf{x}_{t-1} \mid d) \cdot |\mathbf{H}_t|$. Dividing both sides of this equation by $p(\mathbf{x}_{t-1} \mid d)$ gives

$$p(\hat{\mathbf{z}}_t \mid \mathbf{x}_{t-1}, d) = p(\mathbf{z}_t \mid \mathbf{x}_{t-1}, d) \cdot |\mathbf{H}_t|. \quad (17)$$

Since $p(\mathbf{z}_t \mid \mathbf{z}_{t-1}, d) = p(\mathbf{z}_t \mid \mathbf{g}(\mathbf{z}_{t-1}), d) = p(\mathbf{z}_t \mid \mathbf{x}_{t-1}, d)$ and similarly $p(\hat{\mathbf{z}}_t \mid \hat{\mathbf{z}}_{t-1}, d) = p(\hat{\mathbf{z}}_t \mid \mathbf{x}_{t-1}, d)$, Eq. 25 tells us

$$\log p(\hat{\mathbf{z}}_t \mid \hat{\mathbf{z}}_{t-1}, d) = \log p(\mathbf{z}_t \mid \mathbf{z}_{t-1}, d) + \log |\mathbf{H}_t| = \sum_{k=1}^n \eta_{tk}(d) + \log |\mathbf{H}_t|. \quad (18)$$

Its partial derivative w.r.t. $\hat{z}_{ti}$ is

$$\frac{\partial \log p(\hat{\mathbf{z}}_t \mid \hat{\mathbf{z}}_{t-1}, d)}{\partial \hat{z}_{ti}} = \sum_{k=1}^n \frac{\partial \eta_{tk}(d)}{\partial z_{tk}} \cdot \mathbf{H}_{kit} + \frac{\partial \log |\mathbf{H}_t|}{\partial \hat{z}_{ti}} \quad (19)$$

Its second-order cross-derivative is:

$$\frac{\partial^2 \log p(\hat{\mathbf{z}}_t \mid \hat{\mathbf{z}}_{t-1}, d)}{\partial \hat{z}_{ti} \partial \hat{z}_{tj}} = \sum_{k=1}^n \left(\frac{\partial^2 \eta_{tk}(d)}{\partial z_{tk}^2} \cdot \mathbf{H}_{kit} \mathbf{H}_{kjt} + \frac{\partial \eta_{tk}(d)}{\partial z_{tk}} \cdot \frac{\partial \mathbf{H}_{kit}}{\partial \hat{z}_{tj}} \right) + \frac{\partial^2 \log |\mathbf{H}_t|}{\partial \hat{z}_{ti} \partial \hat{z}_{tj}}. \quad (20)$$

The above quantity is always zero according to Eq. 16. Therefore, for each $l = 1, 2, \dots, n$ and each value $z_{t-1,l}$, its partial derivative w.r.t. $z_{t-1,l}$ is always zero. That is,

$$\frac{\partial^3 \log p(\hat{\mathbf{z}}_t \mid \hat{\mathbf{z}}_{t-1}, d)}{\partial \hat{z}_{ti} \partial \hat{z}_{tj} \partial z_{t-1,l}} = \sum_{k=1}^n \left(\frac{\partial^3 \eta_{tk}(d)}{\partial z_{tk}^2 \partial z_{t-1,l}} \cdot \mathbf{H}_{kit} \mathbf{H}_{kjt} + \frac{\partial^2 \eta_{tk}(d)}{\partial z_{tk} \partial z_{t-1,l}} \cdot \frac{\partial \mathbf{H}_{kit}}{\partial \hat{z}_{tj}} \right) \equiv 0. \quad (21)$$

We have made use of the fact that entries of $\mathbf{H}_t$ do not depend on $z_{t-1,l}$. Using different values for d in Eq. 20, take the difference of this equation across them:

$$\begin{aligned} & \frac{\partial^2 \log p(\hat{\mathbf{z}}_t \mid \hat{\mathbf{z}}_{t-1}; r+1)}{\partial \hat{z}_{ti} \partial \hat{z}_{tj}} - \frac{\partial^2 \log p(\hat{\mathbf{z}}_t \mid \hat{\mathbf{z}}_{t-1}; r)}{\partial \hat{z}_{ti} \partial \hat{z}_{tj}} \\ &= \sum_{k=1}^n \left[\left(\frac{\partial^2 \eta_{tk}(r+1)}{\partial z_{tk}^2} - \frac{\partial^2 \eta_{tk}(r)}{\partial z_{tk}^2} \right) \cdot \mathbf{H}_{kit} \mathbf{H}_{kjt} + \left(\frac{\partial \eta_{tk}(r+1)}{\partial z_{tk}} - \frac{\partial \eta_{tk}(r)}{\partial z_{tk}} \right) \cdot \frac{\partial \mathbf{H}_{kit}}{\partial \hat{z}_{tj}} \right] \equiv 0 \end{aligned} \quad (22)$$

where $r \in \{1, 2, \dots, M\}$. If for any value of $\mathbf{z}_t$, the vectors $\mathbf{s}_{t1}, \hat{\mathbf{s}}_{t1}, \mathbf{s}_{t2}, \hat{\mathbf{s}}_{t2}, \dots, \mathbf{s}_{tn}, \hat{\mathbf{s}}_{tn}$ are linearly independent, to make the above equation hold true, one has to set

$$\mathbf{H}_{kit} \mathbf{H}_{kjt} = 0 \quad \text{for } i \neq j.$$

That is, in each row of $\mathbf{H}_t$, there is only one non-zero entry. Since h is invertible, then $\mathbf{z}_t$ must be an invertible, component-wise transformation of a permuted version of $\hat{\mathbf{z}}_t$. So far, the identifiability result has been established.

A.3 Deviations of Transition Prior

Let $\{m_i\}_{i=1,2,\dots,N}$ be a set of learned inverse transition functions that take the estimated latent causal variables, and output the noise terms, i.e., $\hat{\epsilon}_{t,i} = m_i(\hat{z}_{t,i}, \{\hat{z}_{t-\tau}\}_{\tau=1}^L, d)$, where L is the maximum lag. Then we design a transformation $\mathcal{T}(\cdot)$ that maps $[\hat{\mathbf{z}}_{t-L}, \dots, \hat{\mathbf{z}}_{t-1}, \hat{\mathbf{z}}_t]^\top$ to $[\hat{\mathbf{z}}_{t-L}, \dots, \hat{\mathbf{z}}_{t-1}, \hat{\epsilon}_t]^\top$, then the Jacobian matrix of this transformation is:

$$\mathbf{J}_{\mathcal{T}} = \begin{bmatrix} \mathbf{I}_{N \times L} & 0 \\ * & \text{diag}\left(\frac{\partial m_i}{\partial \hat{z}_{t,i}}\right) \end{bmatrix}. \quad (23)$$

Since the matrix is lower triangular, its determinant can be calculated as $|\mathbf{J}| = \prod_{i=1}^N \left| \frac{\partial m_i}{\partial \hat{z}_{t,i}} \right|$. According to the probability density transformation formula, we can get:

$$p(\{\hat{\mathbf{z}}_{t-\tau}\}_{\tau=1}^L, \hat{\mathbf{z}}_t \mid d) = p(\{\hat{\mathbf{z}}_{t-\tau}\}_{\tau=1}^L, \hat{\epsilon}_t \mid d) \cdot \prod_{i=1}^N \left| \frac{\partial m_i}{\partial \hat{z}_{t,i}} \right| \quad (24)$$

Divide both sides by $p(\{\hat{\mathbf{z}}_{t-\tau}\}_{\tau=1}^L \mid d)$ to get:

$$p(\hat{\mathbf{z}}_t \mid \{\hat{\mathbf{z}}_{t-\tau}\}_{\tau=1}^L, d) = p(\hat{\epsilon}_t \mid \{\hat{\mathbf{z}}_{t-\tau}\}_{\tau=1}^L, d) \cdot \prod_{i=1}^N \left| \frac{\partial m_i}{\partial \hat{z}_{t,i}} \right| \quad (25)$$

Therefore, we have:

$$\begin{aligned} \log p(\hat{\mathbf{z}}_t \mid \{\hat{\mathbf{z}}_{t-\tau}\}_{\tau=1}^L, d) &= \log p(\hat{\epsilon}_t \mid \{\hat{\mathbf{z}}_{t-\tau}\}_{\tau=1}^L, d) + \sum_{i=1}^n \log \left| \frac{\partial m_i}{\partial \hat{z}_{t,i}} \right| \\ &= \sum_{i=1}^N \log p(\hat{\epsilon}_{t,i} \mid d) + \sum_{i=1}^N \log \left| \frac{\partial m_i}{\partial \hat{z}_{t,i}} \right| \end{aligned} \quad (26)$$

Since the prior of $p(\hat{\mathbf{z}}_{1:T} \mid d) = p(\hat{\mathbf{z}}_{1:L} \mid d) \prod_{t=L+1}^T p(\hat{\mathbf{z}}_t \mid \{\hat{\mathbf{z}}_{t-\tau}\}_{\tau=1}^L, d)$, we can estimate $\log p(\hat{\mathbf{z}}_{1:T})$ as follows:

$$\log p(\hat{\mathbf{z}}_{1:T} \mid d) = \log p(\hat{\mathbf{z}}_{1:L} \mid d) + \sum_{t=L+1}^T \left(\sum_{i=1}^N \log p(\hat{\epsilon}_{t,i} \mid d) + \sum_{i=1}^n \log \left| \frac{\partial m_i}{\partial \hat{z}_{t,i}} \right| \right) \quad (27)$$

Where the noise $p(\hat{\epsilon}_{t,i})$ and the initial latent variables $p(\hat{\mathbf{z}}_{1:L} \mid d)$ are both assumed to follow Gaussian distributions.

B Datasets

B.1 Datasets Information

PAMAP Dataset. The PAMAP dataset [6] was completed by 9 subjects wearing 3 inertial measurement units and a heart rate monitor. The dataset contains data from 18 different physical activities (e.g., walking, cycling, etc.), where each subject performs 12 of these activities and some subjects perform an additional 6 activities.

USC-HAD Dataset. The USC-HAD dataset [7] is a widely used dataset for Human Activity Recognition (HAR) research. It captures 12 basic daily activities performed by 14 subjects, with each activity repeated 5 times per subject on different days at various indoor and outdoor locations. The dataset is collected using a high-precision MotionNode IMU sensor worn on the front right hip of the subjects, sampling data at 100Hz to capture fine-grained motion dynamics.

WESAD Dataset. The WESAD dataset [8] is a publicly available dataset designed for stress and emotion recognition research. This dataset was collected from 15 participants via sensors worn on the wrist and chest. It contains multiple data modalities such as blood volume pulse, electrocardiogram, respiration, body temperature, three-axis acceleration, etc.

DSADS Dataset The DSADS dataset [9] is designed for human activity recognition (HAR) research, capturing a wide range of daily and sports-related movements. It includes 19 activities (e.g., running, basketball dribbling, rowing) performed by 8 subjects (4 female, 4 male, aged 20 to 30). The sampling rate is 25Hz and the total signal duration is 5 minutes for each activity of each subject.

B.2 Domain Splitting

Although our method does not require manual specification of domain labels, existing OOD generalization methods usually do so. Therefore, in order to achieve a fair comparison with the baseline methods, we randomly divide the dataset into four domains and try to ensure that the number of samples in different domains is balanced after the division. The subject numbering is consistent with the original dataset, and the domain labels are numbered starting from 0. Domain splitting results are shown in Table S1.

Table S1: Initial domain splits.

Domain	0	1	2	3
PAMAP	1	2	5	8
USC-HAD	1, 2, 3, 4	5, 6, 7, 8	9, 10, 11	12, 13, 14
WESAD	2, 3, 4, 5	6, 7, 8, 9	10, 11, 13, 14	15, 16, 17
DSADS	1, 2	3, 4	5, 6	7, 8

B.3 Data Pre-processing

In this section, we introduce the details of data preprocessing. We use a sliding window technique similar to [10] to divide the original time series into multiple data segments. For the WESAD dataset, we set the window size to 200 and the step size to 100, then normalize the data with $\tilde{x} = \frac{x - \min X}{\max X - \min X}$, following Lu et al. [10]. For the PAMAP and USC-HAD datasets, We use the same setting as [11], set the window size to 128 and the step size to 64, then normalize the data with $\tilde{x} = \frac{x - \mu}{\sigma}$, where μ and σ represent the mean and the standard deviation of the dataset, respectively. For the DSADS dataset, we directly utilize the data split by the providers. The time series length, number of input channels and the number of categories of each dataset are shown in Table S2. For evaluation, we use the standard leave-one-domain-out setting [12], where the data from one domain serves as the test set to simulate the OOD distribution, and the

data from the remaining three domains are randomly shuffled and divided into the training and validation sets in the ratio of 8:2. We use the classification accuracy to evaluate the performance of models.

Table S2: Detailed information of datasets.

Dataset	Length	Dimension	Classes
PAMAP	128	30	12
USC-HAD	128	6	12
WESAD	200	8	3
DSADS	125	45	19

C Implementation details

C.1 Network architecture

Our approach consists of an Encoder, a Decoder, a prior network, a causal feature extractor, a domain feature extractor, and a classifier. For the implementation of the Encoder and Decoder, we adopt structures that are similar to [4]. That is, both the encoder and decoder contain five fully connected layers, except for the last fully connected layer, each of which is followed by a LeakyReLU activation function and a dropout layer. For the prior network, we use a two-layer MLP to approximate the transition function $h_i(\cdot)$. For the causal featurizer and domain featurizer, we use a CNN network similar to [10]. The featurizer contains three blocks, each with a 1D convolutional layer, a batch normalization layer, a maxpooling layer, and a ReLU activation function. Finally, we use a single-layer MLP as the classifier. For baseline methods, we adopted the same encoder, featurizer, and classifier to ensure a fair comparison. Note that as a general framework, our approach is not limited to the above network implementations, i.e., it is possible to implement each of the above modules using state-of-the-art architectures to achieve stronger OOD generalization capabilities.

C.2 Model training and parameters

In this section, we will introduce the training and testing process of the model as well as the parameter configurations. We use a two-stage training process for iterative optimization. The process is divided into the causal prediction stage and the domain discovery stage. In the causal prediction stage, we perform class prediction and optimize the causal graph, the causal representation learning network, the causal featurizer and the classifier. In the domain discovery stage, we perform domain label prediction and only optimize the causal graph and the domain featurizer. At inference time, we only use the discovered causal graph, encoder, feature extractor, and classifier to obtain predictions. Therefore, no additional computational burden is introduced in the testing phase compared to the baseline methods.

For the WESAD dataset, we set the batch size to 64 and the maximum training epoch to 50, and for the other datasets we set the batch size to 32 and the maximum training epoch to 150. Optimization was performed using the AdamW optimizer with a weight decay of 1×10^{-4} . We set the learning rate to 1×10^{-3} for the sequential VAE module and 1×10^{-4} for the other

modules. Other hyperparameters such as λ and K are carefully tuned based on the task. All the experiments are implemented with PyTorch [13] on an NVIDIA RTX 3090 24GB GPU.

D Additional experiments

D.1 $\mathcal{H}$ -divergence visualization

In this section, we set the number of domains $K = 3$ and report the $\mathcal{H}$ -divergence between domains to quantify their distributional differences. As shown in Figure S1, compared with initial random splits and splits from the Diversity method, our prototype-based domain inference algorithm yields significantly higher $\mathcal{H}$ -divergence between domains, which means that our method can uncover more diverse domains. This diversity encourages the identification of latent variables and enhances the learning of causally invariant representations.

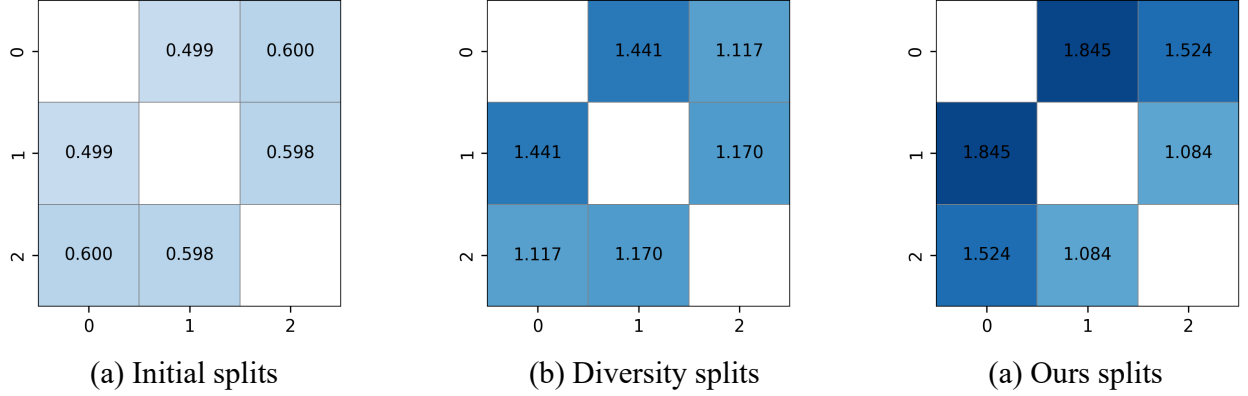


Figure S1: $\mathcal{H}$ -divergence among domains on PAMAP dataset. Axes are domain numbers.

D.2 Sensitivity analysis

In this section, we show sensitivity analysis results of the hyperparameters K and λ in Figure S2. The hyperparameter K represents the number of potential domains. When K is small, data from different domains may be mistakenly assigned to the same domain, and the learned invariant representation may retain spurious domain information, impairing the OOD generalization ability. When K increases beyond the actual number of domains, the optimization of the latent variable identification network and domain prototypes becomes more challenging. The hyperparameter λ controls the sparsity of the causal graph. When λ is relatively small, some non-causal factors will also be used for OOD prediction, thereby introducing spurious associations; when λ is large, some causal factors will be discarded, thereby violating the causal sufficiency principle and causing OOD generalization performance to deteriorate.

D.3 Convergence analysis

The convergence analysis results of the method are shown in Figure S3. From the training loss curve, it can be seen that although there are fluctuations in the training process, the overall loss is constantly decreasing and eventually converges. In addition, Figure S3 (b) is the visualization of the discovered causal probability graph $\mathcal{G}$, where each element represents a latent variable,

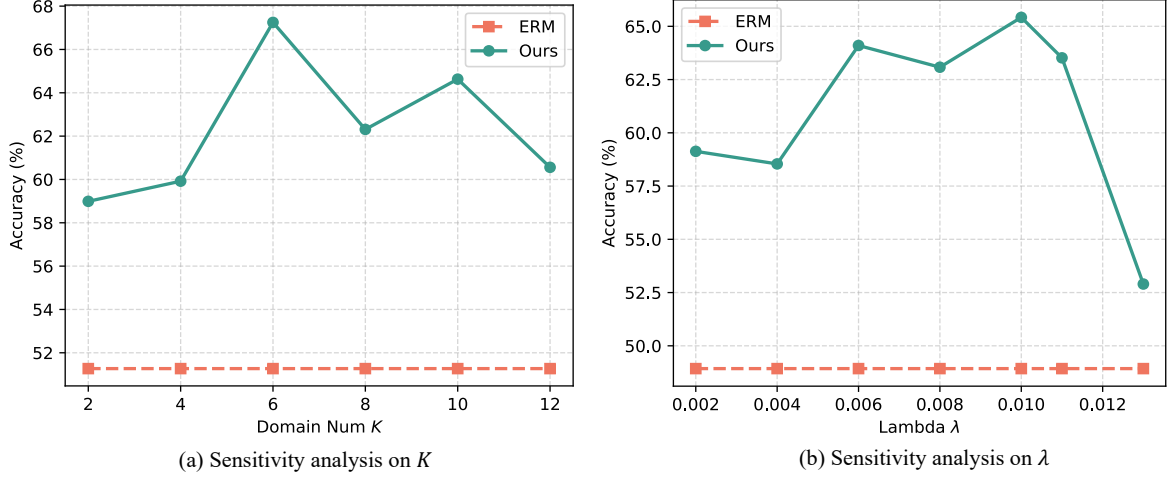


Figure S2: Results of sensitivity analysis on PAMAP dataset.

and the value represents the probability that the latent variable is the cause of Y . It can be seen that after two-stage alternating optimization, our learnable causal mask can converge to a stable causal graph that can distinguish causal variables from non-causal variables.

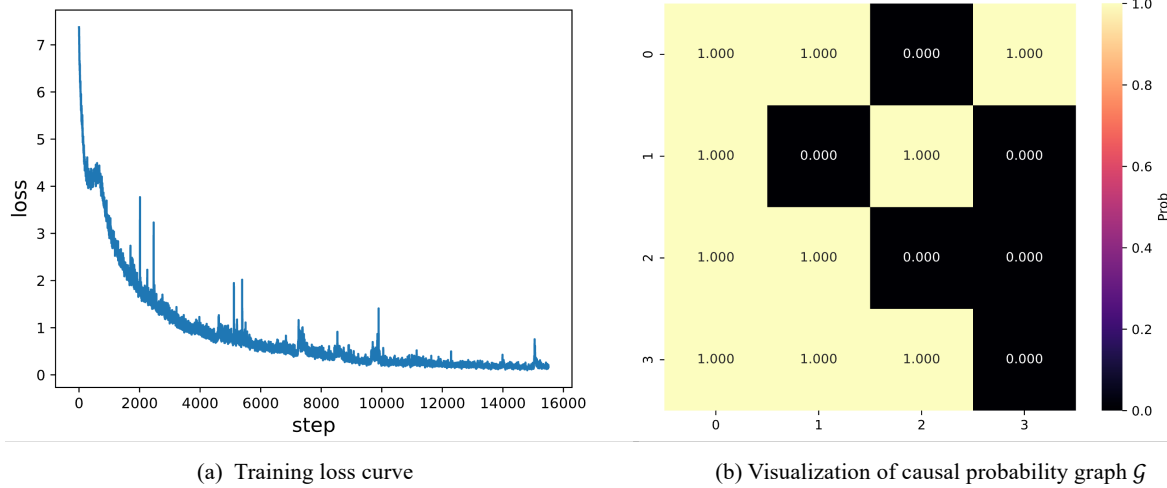


Figure S3: Results of convergence analysis on WESAD dataset.

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